

Study on the wind load and wind-induced interference effect of photovoltaic (PV) arrays on two-dimensional hillsides

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ABSTRACT

Accurate assessment of wind loads on PV modules is crucial for the economic efficiency and safety of PV power stations. Most of these studies focused on the PV arrays installed on flat ground, whereas research on the PV arrays installed on hillsides has been lacking. This paper carried out CFD simulations of single-row PV modules and arrays on a two-dimensional hillside. The results show that the slope can either enhance or weaken the wind load. The enhancement and weakening effects become stronger with larger slope. When the slope is 30°, the wind load of the single row of PV modules at the bottom of the hillside can be reduced by up to 80%, the load of the first row of the array can be reduced by up to 25%, the load of the single row of the PV modules at the top of the hillside can be enhanced by up to 150%, and the load of the last row of the array can be enhanced by up to 280%. The wind field in the range of 0.2 times the hillside length from the top of the hillside is more complex, and the wind load is quite different from that of the flat ground, which should receive special attention in the design process.

1. Introduction

Tracking PV systems enhance power generation by adjusting module tilt angles, offering efficiency improvements of 10–30 % compared to those of fixed systems. [4,13,15,31] Single-axis PV trackers are widely used because they ensure power generation efficiency at a lower cost than dual-axis PV systems. [4] The rotational inertia of single-axis trackers is provided by the PV modules, while the rotational stiffness is solely provided by the axis, resulting in a low frequency and a 'slender and flexible' characteristic. This renders single-axis trackers wind-sensitive and prone to damage in strong winds. [29] Therefore, the design of wind-resistant single-axis trackers is challenging.

To address these concerns, many studies have explored the effects of wind loads on PV modules. The tilt angle and wind direction significantly influence the wind load on single-row PV modules. The tilt angle directly impacts the strength and distribution of wind loads on PV modules. The tilt angle leads to an uneven wind load distribution along the chord length, subjecting PV modules to torque under wind pressure.

[22] Research [1,9] using wind tunnel experiments and computational fluid dynamics (CFD) numerical simulations revealed that larger tilt angles increase the wind load, while smaller tilt angles reduce it. The wind direction alters the wind load distribution on the modules, significantly impacting the load at the edges. Using wind tunnel experiments, [21] observed greater wind loads at PV module edges under oblique winds, with the maximum load in the perpendicular direction. The torque on the PV modules was found to vary significantly with the wind direction angle. [10,11].

In contrast to wing models that are unaffected by the ground, PV modules experience significant ground effects. Consequently, several studies have examined how ground clearance affects the wind load on these modules. Increased ground clearance weakens the ground effect, resulting in stronger vortex shedding. Decreased ground clearance strengthens the ground effect, causing irregular vortex shedding. Such irregular vortex shedding makes the wind load on PV modules more complex. [25,26].

The wind load on the PV arrays shows significant interference due to marked flow pattern changes after the wind passes through the modules,

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Nomenclature			
s	Row spacing, (m)	E	Empirical constants of smooth walls
$\Delta\beta$	Relative tilt angle, ($^{\circ}$)	z^*	Dimensionless distance
B	Tilt angle, ($^{\circ}$)	C_{μ}	Empirical coefficient
θ	Topography slope angle, ($^{\circ}$)	ΔB	Correction constant of standard wall function
θ_w	Angle between wind direction and horizontal plane, ($^{\circ}$)	$Y1$	The length before the hillside, (m)
$\Delta\theta$	The relative angle between the wind direction and the hillside, ($^{\circ}$)	$Y2$	The horizontal projection length of the hillside, (m)
U	Wind speed, (m/s)	$Y3$	The length after the hillside, (m)
u^*	Friction speed, (m/s)	C_M	The average torque coefficient
K	Von Carmen's constant	C_p	The average pressure coefficient
z_0	Roughness length, (m)	ρ	The air density, (kg/m^3)
C_1	The fitting constants of Eq. (2) and Eq. (3)	U_h	The inflow wind speed at the height of the PV module's rotation axis above flat ground, (m/s)
C_2	The fitting constants of Eq. (2) and Eq. (3)	C	The chord length of the PV modules, (m)
K	Turbulent Kinetic Energy, (m^2/s^2)	L	The span dimension of the PV modules, (m)
ε	Turbulent Dissipation Rate, (m^2/s^3)	L_h	Length of the hillside, (m)
U^*	Dimensionless wind speed, (m/s)	y_i	The distance from point i on the hillside to the bottom of the hillside, (m)

making the load on the internal modules more complex. In addition to the tilt angle, wind direction, and ground clearance, row interference notably impacts the wind load. The first row of an array experiences a slightly weaker wind load than a single row, with the tilt angle variation affecting the wind load in a similar manner [18]. The first row of PV modules experiences the maximum wind load, with a significant reduction in the wind load for the second row due to the shielding effect of the front row. A smaller reduction in the wind load is obtained for the third and fourth rows [32]. A previous study [5] revealed that peak torque under high-speed wind was higher than that predicted from pressure data, motivating the proposal of a new method for assessing wind loads on PV arrays. Wind tunnel test results show that the fluctuating wind load on PV arrays follows a different pattern from the static wind load. A tilt angle of 15° serves as a dividing line: below this angle, a shielding effect is observed, and above this angle, an amplification effect occurs. [18] Row spacing in PV arrays significantly influences land utilization, light collection rates, and wind load. Increasing the row spacing changes the leeward flow field pattern of PV modules, leading to an increase in the wind load and a decrease in the shielding effect. [26].

An accurate wind load assessment for single-axis trackers can substantially reduce costs. [14] indicated that wind loads at PV array edges can be more than two times greater than those at the interior positions. This necessitates zoning wind loads within the arrays. [18] defined the perimeter as the area extending seven chord lengths from the edge of the PV modules, considering the rest as the interior. This zoning approach can lead to significant cost savings.

Flow separation, attachment, and vortex shedding occur when wind flows through a PV module, giving rise to complex aerodynamic forces [8]. For the PV modules, the stiffness decreases with increasing span. When the stiffness is large, the wind load is equivalent to the static force, and the wind-induced vibration phenomenon is almost entirely absent. When the stiffness is low, the PV module is affected by both static and dynamic forces, and wind-induced vibration may occur. Factors such as the tilt angle, torsional stiffness, chord length, moment of inertia, and natural frequency may affect the wind-induced vibration of PV modules. The tilt angle is the main factor affecting the critical wind speed. In previous studies [19,20,24], the wind-induced torsional vibration of a single axis was studied via wind tunnel tests and numerical simulations. It was found that the critical wind speed varied strongly with tilt angle. In [20], the wind-induced torsional vibration of a flat single shaft was studied via a wind tunnel test. The critical wind speed was greater near a tilt angle of 0° , and was lower in the tilt angle ranges of $-20^{\circ} \sim -15^{\circ}$ and $20^{\circ} \sim 30^{\circ}$. When the tilt angle was less than -45° or greater than 55° , the wind-induced torsional vibration phenomenon was not observed. The

torsional stiffness, chord length, moment of inertia and natural frequency are the secondary factors that affect the critical wind speed. [19] studied these parameters and reported that the variation of the wind speed with the inclination angle was small under different parameters.

While extensive research has been conducted on PV array wind loads, offering valuable insights for engineering, studies specifically targeting hillside PV arrays are lacking. In engineering practice, an ideal flat ground does not exist, wind is influenced by any object that it flows over, and hillsides can greatly change the characteristics of wind flow. Hillside wind fields are complex and feature convergent, separated, and reverse flows. Studies of wind fields over typical hillsides has significant practical value. Research on wind fields over typical two-dimensional [6,17] and three-dimensional hillsides [16,34] has yielded meaningful conclusions. Moreover, standards adopted in various countries describe the terrain acceleration effect on wind fields in detail. While these conclusions and standards provide some general guidance for determining wind load values for PV arrays on hillsides, the evaluation of the wind load values for these PV arrays is not specifically address by current research and standards. Hillsides change the wind direction and speed. Hence, when PV arrays are located on hillsides, it is highly important to accurately estimate the flow fields around PV modules and to take these flow fields into account in the design.

The present study employs numerical simulations to determine the wind load values and interference effects on PV arrays on hillsides, assessing how slope, location, tilt angle, and interference impact wind loads. Section 2 describes the numerical methods, including boundary conditions, model dimensions, and mesh details. Section 3 validates the accuracy of the numerical methods and analyzes the results. In particular, Section 3.1 simulates and analyzes the wind environment around PV modules, focusing on wind direction and speed. Section 3.2 analyzes the wind loads on single-row PV modules at different slopes and locations. In Section 3.3, appropriate locations for PV arrays on hillsides are selected based on the calculation results from Section 3.2, and the impact of the slope on the shielding effect of the wind loads on PV arrays is analyzed. Section 3.4 analyzes the pressure coefficients and velocity contours under typical conditions and studies the mechanism of the effect of the slopes on the wind load on PV modules.

2. Methodology

This section introduces the details of the numerical simulation, including the size of the computational domain, the sensitivity analysis of the number of meshes, and the parameter settings for the boundary conditions.

2.1. Computational domain and the mesh scheme

In this study, a two-dimensional hillside is used for modeling. This two-dimensional hillside is similar to the roofs with different slopes examined in previous studies [2]. However, the two-dimensional hillside examined in this study differs from the roof in that the PV module is set at a certain angle relative to the hillside. Although an ideal two-dimensional hillside is never found in engineering practice, this study is necessary. The results of this study show that the wind load of PV arrays installed on hillsides is different from that of arrays installed on flat ground; this information can then be used by PV designers. This study also provides reference information for future research on PV arrays on three-dimensional hillsides and real complex mountains. For example, [7,12] studied the wind field of two-dimensional hillsides, obtained conclusions to guide practical engineering, and provided basic theoretical guidance for three-dimensional mountains and complex mountain wind fields.

Fig. 1 illustrates the computational domain with the size of $20L(X) \times 200s(Y) \times 26L(Z)$ that follows the AIJ (Architectural Institute of Japan) guidelines for computational domain dimensions [28]. Here, $L=15$ m is the span of the PV modules, and $s=5.64$ m is the row spacing within the PV array. In this study $Y1$ is defined as the length before the hillside, $Y2$ as the hillside's horizontal projection, and $Y3$ as the posthillside length, with a ratio of $Y3/Y2 > 7$. According to the literature [6], a $Y3/Y2$ ratio > 7 suggests an effectively infinite hillside top that minimizes the flow disturbances from a short hill crest. A polyhedral mesh is used to discretize the computational domain and the PV module model, with local refinement of the surface mesh of the PV modules and boundary layers set on the walls to capture the near-wall flow.

2.2. Numerical calculation method and parameter setting

This paper employs a three-dimensional steady-state Reynolds Average Navier-Stokes (RANS) method for simulating wind loads on PV arrays using a pressure-based solver. The realizable $k-\varepsilon$ turbulence model is used in the simulation [27], with velocity-pressure coupling handled by the SIMPLEC [30]. The nonlinear discretization scheme for the turbulence model and momentum equation employs a second-order upwind scheme. For all involved physical quantities, the residual criterion is set to below 1×10^{-5} .

2.3. Boundary condition setting

The computational domain inlet is set as a velocity inlet; the outlet is

set as a pressure outlet; both the top and sides of the computational domain are set as symmetric; and the surfaces of the PV module model and the bottom of the computational domain are set as walls.

Accurate simulation of the atmospheric boundary layer is a prerequisite for ensuring the accuracy of numerical simulation results. Based on the inlet boundary mathematical model (RH model) proposed by Richards and Hoxey [23], a new set of mathematical models was derived by Yang et al. [35]. The turbulence energy of this model changes with height, obtaining closer agreement with the measurement results. This model is derived from the standard $k-\varepsilon$ turbulence model. Xu et al. [33] combined this model with the realizable $k-\varepsilon$ turbulence model and made corrections with wall functions, showing good self-sustainable effects. This study adopts the inflow conditions used by Xu et al. [33] as the inlet boundary conditions for this paper, as shown in the following equation:

$$U = \frac{u_*}{\kappa} \ln \frac{z + z_0}{z_0} \quad (1)$$

$$k = \frac{u_*^2}{\sqrt{C_\mu}} \sqrt{C_1 \ln \left(\frac{z + z_0}{z_0} \right) + C_2} \quad (2)$$

$$\varepsilon = \frac{u_*^3}{\kappa(z + z_0)} \sqrt{C_1 \ln \left(\frac{z + z_0}{z_0} \right) + C_2} \quad (3)$$

$$U^* = \frac{1}{\kappa} \ln(Ez^*) - \Delta B \quad (4)$$

Where κ is von Carmen's constant, u_* is the friction speed, z_0 is the roughness length, and C_1 and C_2 are constants. ΔB is the correction constant for the wall function. The values of the related parameters are listed in Table 1.

The inflow conditions used in the numerical calculations of this study were obtained by measurements performed during wind tunnel tests and were fitted by the above formula. The wind tunnel test is a rigid pressure measurement test of a PV array. It is carried out in the low-speed section (length, width and height of 24, 4, and 3 m, respectively) of the wind tunnel laboratory of Shijiazhuang Tiedao University. The atmospheric

Table 1

Parameters related to the inlet wind speed profile and turbulence characteristics profile.

κ	u_*	z_0	C_μ	C_1	C_2	ΔB
0.42	0.39898	1.25644×10^{-4}	0.047	-0.8364	1.25053	1.5

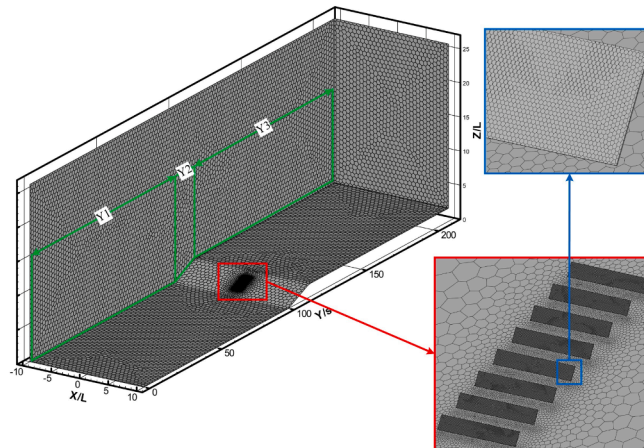


Fig. 1. Mesh of computational domain and PV module.

boundary layer wind field is simulated by rough elements and gratings. A photograph of the test equipment is shown in Fig. 2.

The measured data and fitting curves of the wind tunnel test are shown in Fig. 3. The application of a mathematical model as an inlet condition can achieve a better self-sustainable effect, as shown in Fig. 4. This figure shows the results of a trial calculation in a domain without obstacles. The relevant parameters obtained by fitting are listed in Table 1.

2.4. Definitions of the main parameters

$$C_p = \frac{\sum_{i=1}^N P_i A_i}{1/2 \rho U_h^2 L C} \quad (5)$$

$$C_M = \frac{M}{1/2 \rho U_h^2 L C^2} \quad (6)$$

Where C_p and C_M are the average pressure coefficient and the average torque coefficient, respectively. P_i is the pressure of each grid of the PV module numerical model, A_i is the area of each grid, N is the number of surface grids, M is the torque of the PV module (the torsion center is the geometric center of the PV module model, and M is monitored by the CFD solver), and U_h is the inflow wind speed at the height of the PV module's rotation axis above flat ground. where ρ is the air density, L is the span length of the module, and C is the chord length of the module.

Fig. 5 presents a schematic of the model used for this study, where β is the tilt angle, $\Delta\beta$ is the relative tilt angle, and θ is the slope angle of the topography. The tilt angle β is positive when the upper surface of the PV module faces the wind and is negative when the lower surface faces the wind. h is the height from the ground of the PV module. Geometric characteristics of the PV modules are listed in Table 2.

2.5. Mesh sensitivity analysis

Fig. 1 illustrates a polyhedral mesh division scheme for discretizing the computational domain and model. Three mesh schemes, based on the boundary conditions described in Section 2.2, are utilized for mesh sensitivity analysis. The average pressure coefficients of the PV modules at various locations in the flat PV array serve as comparison parameters, with the results for the three mesh schemes depicted in Fig. 6. While the trends of the pressure coefficients with location remains consistent across the three mesh schemes, their magnitudes vary. A mesh count of 114 W results in higher pressure coefficients for the first two rows. Elevating the mesh count to 215 W decreases these coefficients, and a further increase to 400 W yields no significant changes. Considering the



Fig. 2. Rigid pressure measurement test of PV array.

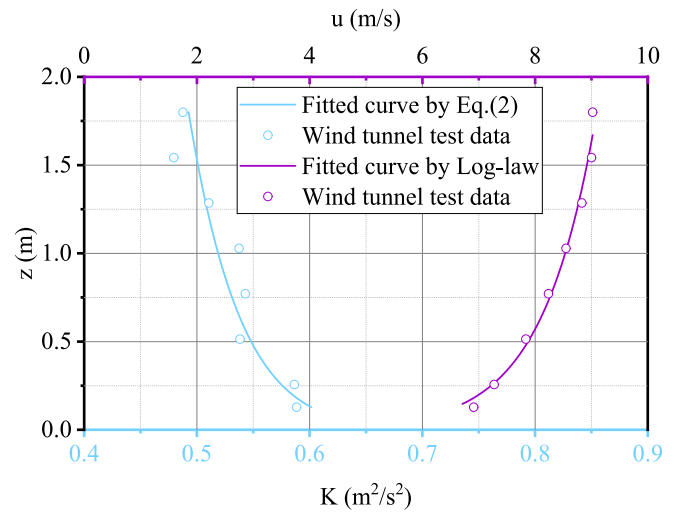


Fig. 3. Wind tunnel measured wind speed and turbulence energy and its fitting curve.

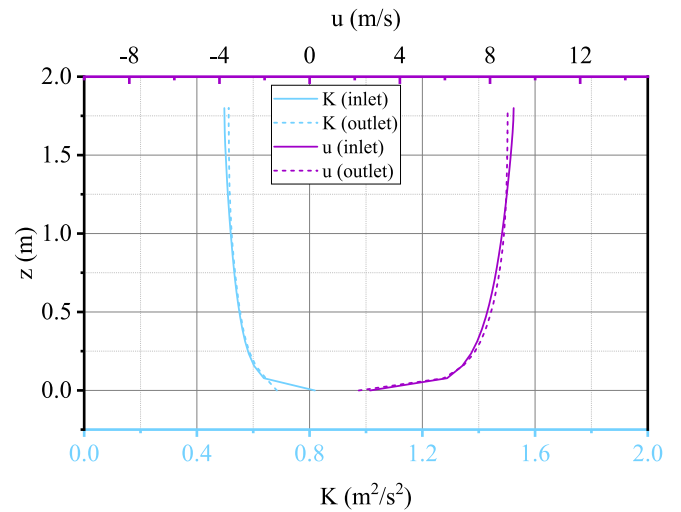


Fig. 4. Wind speed and turbulence energy profiles at the entrance and exit of the domain without obstacles.

computational efficiency and accuracy of the results, this paper selects the 215 W mesh count scheme. The final y^+ values are maintained between 30 and 150, facilitating the use of standard wall functions for near-wall solutions.

3. Results and discussion

This study designates a single row of PV modules as R0, the windward first row as R1, the second row as R2, and rows three through eight as R3-R8. To validate the accuracy of the numerical simulations, this study compared the numerical simulation results for R1 on flat ground to those of wind tunnel tests and other studies, as illustrated in Fig. 7. In Fig. 7 (left), the black circles depict the results obtained in wind tunnel test in this study, with each circle representing the average pressure coefficient along the span. The specific details of the experiment can be found in the literature [18]. The numerical results are in close agreement with the results of the wind tunnel tests and match the trends reported in other studies, despite some minor differences in the pressure coefficients at certain tilt angles. Fig. 7 (right) shows the wind pressure coefficient at different positions in the PV array when the inclination angle is 60° . The figure shows that the difference between the simulation

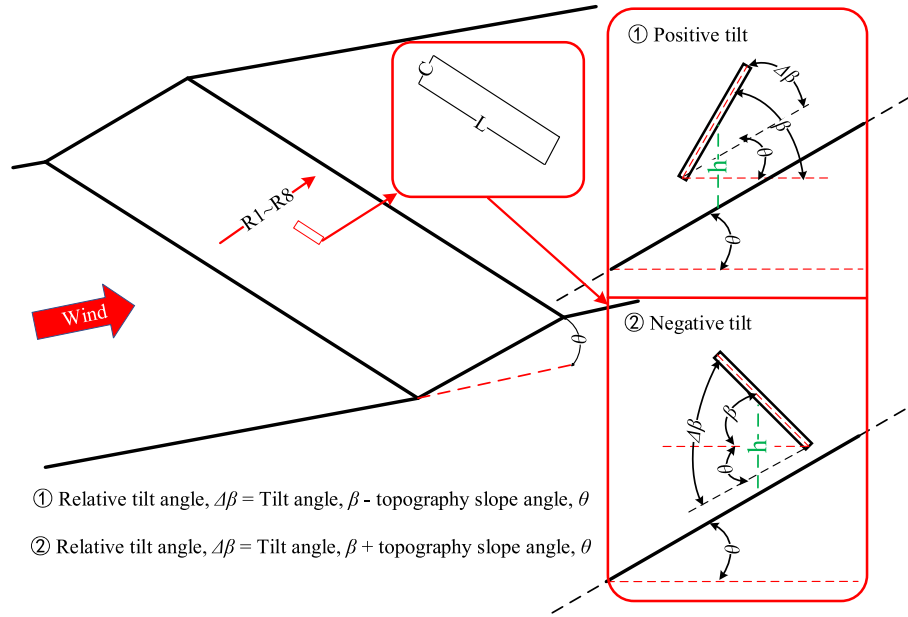


Fig. 5. Model diagram.

Table 2
Geometric characteristics of the PV modules.

s	h	C	L
5.64 m	1.8 m	2.4 m	15 m

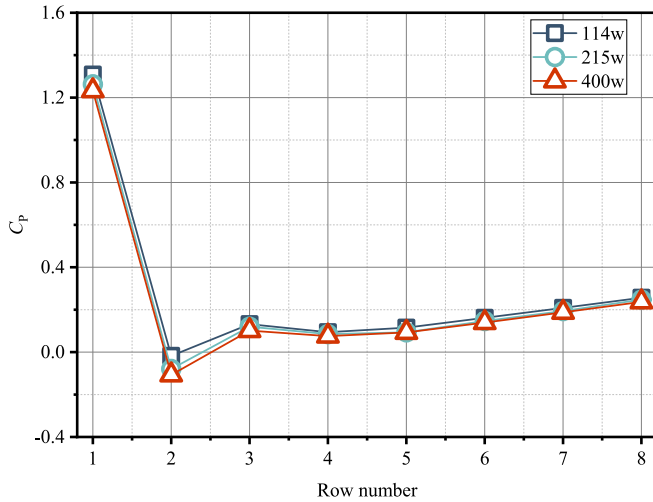


Fig. 6. Average wind pressure coefficients of PV modules at different locations under three mesh schemes.

results of this paper and the results reported in the literature [18] is small, and both sets of results show the same trend. The pressure coefficient of R1 is quite different from the results in the literature [21], mainly because the definition of the reference wind speed is different. In this paper, the wind speed at the rotating shaft is defined as the reference wind speed, while the wind speed at the upper edge of the PV module is defined as the reference wind speed in the literature [21]. A similar study [3] defined the reference wind speed as the wind speed at a height of 10 m. Overall, the numerical calculation scheme of this study can well reflect the variation pattern of the wind loads on PV arrays and can be applied in further research.

Table 3 displays the variables involved in the numerical simulation.

To compare with the wind load on PV modules on a slope, in addition to the values listed in the table, simulations were also performed for $\beta = \pm 90^\circ$, $\pm 80^\circ$, and $\pm 70^\circ$ when $\theta = 0^\circ$. An extensive range of β and $\Delta\beta$ variations is explored that is unlikely to be encountered in real engineering projects. Nonetheless, these calculations were conducted in order to elucidate the relationship between the wind load and $\Delta\beta$.

3.1. Analysis of the wind field on the hillside

Hillside locations alter wind speed and direction. This study assessed how hillside locations affect wind loads by performing wind load simulations for single-row PV modules at five typical locations. Fig. 8 shows the single-row PV module locations on the hillside, where L_h denotes the hillside length.

The wind field characteristics on the hillside are closely linked to the wind load on hillside PV modules. Hence, the authors studied hillside wind field characteristics before analyzing the PV module wind loads. Given the focus of this study, wind speed monitoring was limited to two straight lines. In Fig. 9, the two red lines mark the monitoring locations, ensuring that the PV modules fall within their range at any tilt angle. This study describes wind field characteristics using relative angle $\Delta\theta$, streamwise wind speed ratio R_y , and vertical wind speed ratio R_z . The relative angle between the wind direction and the hillside is defined as $\Delta\theta = \theta - \theta_w$, and R_y and R_z are obtained by normalizing the streamwise and vertical wind speeds to the inflow wind speed at the height of the rotation axis of the PV module on flat ground. R_y and R_z are defined as follows:

$$R_y = \frac{U_y}{U_h} \quad (7)$$

$$R_z = \frac{U_z}{U_h} \quad (8)$$

The wind direction varies with changes in the hillside location. Fig. 10 shows that the angle $\Delta\theta$ between the wind direction and hillside decreases, stabilizes, and then increases from bottom to top, displaying symmetry. The angle $\Delta\theta$ peaks at the hillside's top and bottom and decreases in the middle, highlighting turning angles at both ends as the key factors for changing the wind direction. The wind direction shifts due to the turning angle at the hillside bottom and then gradually becomes parallel to the hillside. Near the hillside's top, the wind direction swiftly

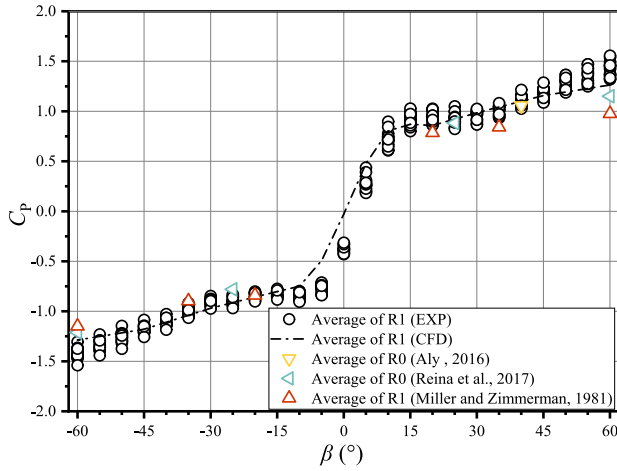
Fig. 7. Average pressure coefficient when $\theta = 0$.

Table 3

List of numerical simulation variables.

β	$\pm 60^\circ, \pm 45^\circ, \pm 30^\circ, \pm 20^\circ, \pm 15^\circ, \pm 10^\circ, \pm 5^\circ, 0^\circ$
θ	$30^\circ, 20^\circ, 15^\circ, 10^\circ, 5^\circ, 0^\circ$

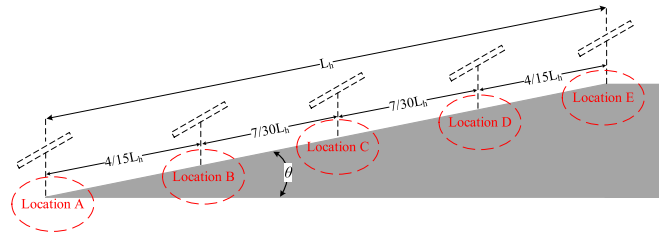


Fig. 8. Location diagram of single-row PV module.

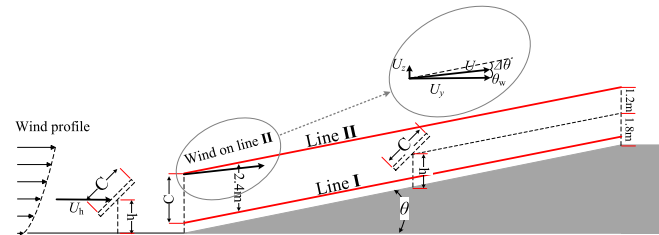
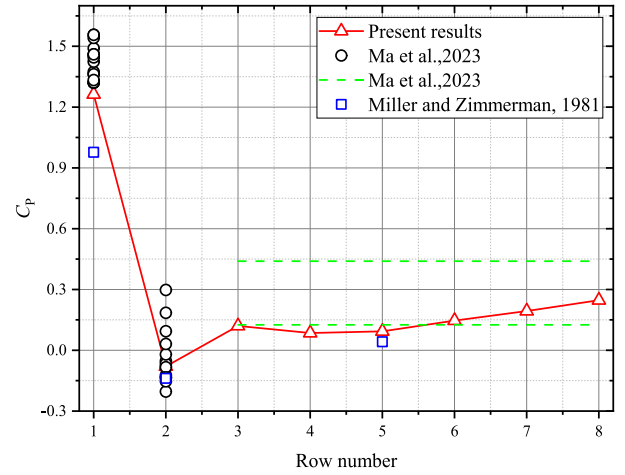


Fig. 9. Diagram of the wind field model on the hillside.

changes because of the top's turning angle and then slowly returns to horizontal without hillside influence. The angles at the hillward top and bottom are nearly identical, indicating equal influence on the change in the wind direction and linking the turning angle to the directional shift capacity.

The characteristics of the wind direction at different heights from the hillside vary. Comparison of the data for Lines I and II in Fig. 10 shows that except for the top and bottom of the hillside, the angle between the wind and the hillside on Line II is generally greater than that on Line I, and the degree of wind direction change on Line II is weaker than that on Line I. This suggests that the influence of the slope on the wind diminishes with increasing distance from the hillside. For $\theta = 5^\circ, 10^\circ$, and 15° , the angles between the wind direction and hillside at the top and bottom of Lines I and II are nearly identical. However, differences in the angle between the wind direction and the hillside at the top of Lines I and II occur when $\theta = 20^\circ$ and 30° . This may be caused by flow



separation due to excessively steep slopes. [12] reported that wind tunnel experiments showed no flow separation on 7.5° and 15° slopes but did observe flow separation on $30^\circ, 45^\circ$, and 60° slopes.

The slope has a significant effect on the wind direction. Comparing Fig. 10(a)–(e), it is observed that the angle between the wind direction and the hillside increases with larger slope, indicating that a larger slope has a stronger influence on the wind direction. At the bottom and top of the hillside, the angle between the wind direction and the hillside is approximately half of that of the slope, i.e., $\Delta\theta \approx 0.5\theta$.

In addition to the wind direction, wind speed is also crucial for the wind load on PV modules. Fig. 11 shows the distribution of streamwise and vertical wind speed ratios at different locations. In the figure, red and blue indicate streamwise and vertical wind speed ratios, respectively, with solid lines for Line I and dashed lines for Line II. The gap between the solid and dashed lines shows the wind speed ratio difference between Line I and Line II. Near $y_i/L_h = 0.8$, the vertical wind speed ratio for Line I is greater than that for Line II, and with steeper slopes, the streamwise wind speed ratio near the slope's top for Line I also exceeds that of Line II.

First, the streamwise wind speed ratio is analyzed. At the hillside's bottom, the streamwise wind speed ratio typically falls below 1, decreasing further with increasing slope steepness. This indicates that the hillside at the bottom acts to reduce the streamwise wind speed. From the bottom to the top of the hillside, the streamwise wind speed ratio gradually increases, and the slope increasingly acts to enhance the streamwise wind speed, with a greater slope leading to a greater rate of increase. The width of the red shaded area gradually decreases from the bottom to the top of the hillside, indicating that the difference in the wind speed between Line I and Line II gradually decreases from the bottom to the top. When the slope is too steep, the streamwise wind speed on Line I can even exceed that on Line II. This indicates that the acceleration effect of the streamwise wind speed increases with closer proximity to the top of the hillside, with the streamwise wind speed at lower locations near the top of the slope even exceeding that at higher locations.

The characteristics of the vertical wind speed ratio differ from those of the streamwise wind speed ratio. Across all five slopes studied, the vertical wind speed ratio remains below 0.8 for all locations from the bottom to the top of the hillside. Overall, from the bottom to the top of the hillside, the vertical wind speed ratio shows a trend of first increasing and then decreasing, with the highest vertical wind speed ratio occurring near $y_i/L_h = 0.9$. Except near the top of the hillside, the vertical wind speed on Line I is less than that on Line II. Near the top of the hillside, the vertical wind speed on Line I exceeds that on Line II. This indicates that the acceleration effect of the vertical wind speed becomes stronger with lower location near the top of the hillside.

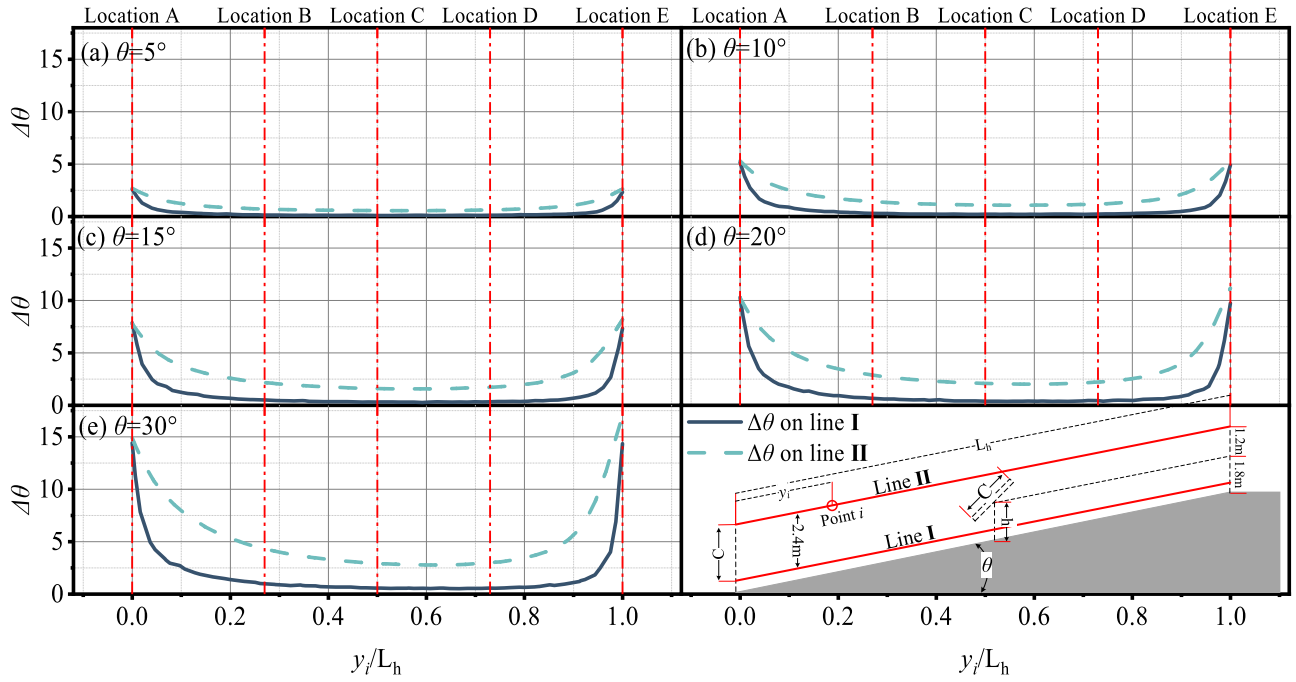


Fig. 10. The angle between the wind direction and the hillside at different locations on the hillside.

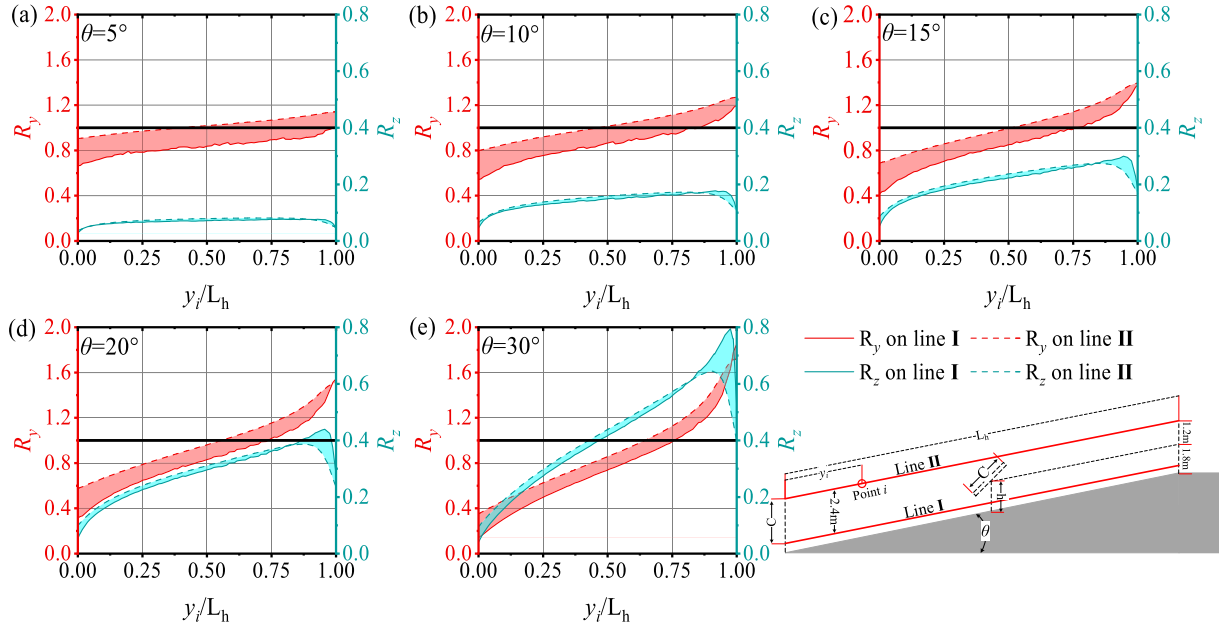


Fig. 11. The ratio of streamwise wind speed to vertical wind speed at different locations on the hillside.

3.2. Wind load analysis of single-row PV modules

Hillside-induced wind direction changes suggest a close relationship between the relative tilt angle $\Delta\beta$ and the wind load on the PV modules. Fig. 12 compares the pressure coefficients for single-row PV modules on hillside and those for the modules on flat ground, with red circles representing the pressure coefficients on flat ground (where $\theta = 0^\circ$ makes $\Delta\beta$ equivalent to β). The figure shows that except for at the top and bottom of the hillside, the trend of the change in the relative tilt angle for the PV modules at other hillside locations is consistent with the trend of the change in the tilt angle for the PV modules on flat ground. The wind load on the PV modules varies significantly between different locations on the hillside. From Location A to Location B, the wind load on the PV

modules on the hillside is less than that on flat ground, indicating that the slope reduces the wind load on the PV modules, and this reduction effect is enhanced with increasing slope. From Location B to Location C, the wind load on the PV modules on the hillside is slightly greater than that on flat ground, but the overall difference is not significant. From Location C to Location E, the wind load on the PV modules on the hillside is greater than that on flat ground, indicating that the slope enhances the wind load on the PV modules, and this enhancement effect becomes stronger with closer proximity to the top of the hillside.

Fig. 13 compares the torque coefficients for single-row PV modules on hillside and on flat ground, with red circles representing the torque coefficients on flat ground (where $\theta = 0^\circ$ makes $\Delta\beta$ equivalent to β). The torque coefficients of the PV modules at different locations have similar

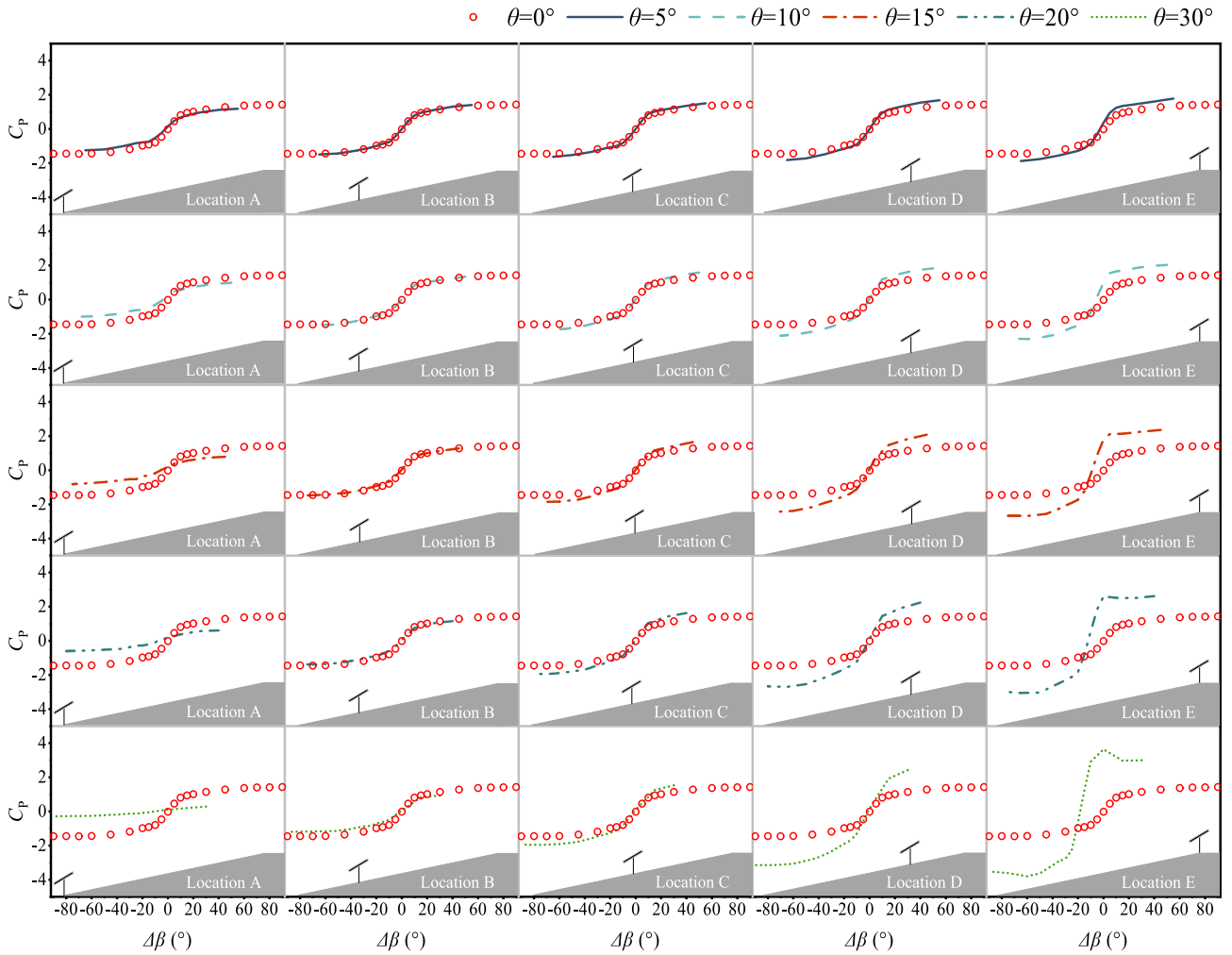


Fig. 12. Comparison of the pressure coefficients of single-row PV modules at different locations on the hillside and on the flat ground.

characteristics to those of the pressure coefficients. From Location A to Location B, the torque coefficients of the PV modules on the hillside are all lower than those for the PV modules on flat ground, indicating that the slope reduces torque, and this reduction effect is enhanced with increasing slope. From Location B to Location C, the torque coefficients of the PV modules on hillsides are slightly greater than those on flat ground, but the overall difference is not significant. From Location C to Location E, the torque coefficients of the PV modules on the slope are greater than those on flat ground, indicating that the slope enhances the torque coefficients of the PV modules, and this enhancement effect becomes stronger closer to the top of the hillside.

To visually compare the pressure and torque coefficients of PV modules at various locations on the same slope, these coefficients are plotted in Figs. 14 and 15, respectively. Both figures show that the pressure and torque coefficients for the PV modules vary across the hillside and increase gradually from bottom to top. The slope influences the pressure and torque coefficients, with steeper slopes giving rise to larger differences in the pressure coefficients at various hillside locations.

Fig. 14 shows the pressure coefficients of single-row PV modules under different slopes. The pressure coefficients on the hillside increase with increasing relative tilt angle. With the exception of the top of the hillside, the pressure coefficients of the PV modules at the other locations exhibit similar characteristics: the pressure coefficients change sign at 0° , meaning that the direction of the pressure is reversed; the curve of the pressure coefficient variation as a function of the relative tilt angle is roughly centrally symmetric, with greater difference in the

pressure coefficients of the PV modules at different locations obtained with larger relative tilt angle. For PV modules located at the top of the hillside, when the slope is smaller, the pressure coefficients also show characteristics similar to those at other locations. As the slope increases, the curves of the pressure coefficient variation as a function of the relative tilt angle no longer exhibit central symmetry.

Fig. 15 shows the torque coefficients of the single-row PV modules under different slopes. The torque coefficients for the PV modules on the hillside reach the maximum values at a relative tilt angle of $\pm 15^\circ$. For locations other than the top of the hillside, the torque coefficients share certain characteristics: they change sign at 0° , indicating torque direction reversal. Their distribution is nearly centrally symmetric around 0° , with the greatest differences occurring at $\pm 15^\circ$. For PV modules located at the top of the hillside, when the slope is smaller, their torque coefficients also exhibit characteristics similar to those at other locations. As the slope increases, the curves of the variation of the torque coefficients as a function of the relative tilt angle no longer exhibit central symmetry.

Fig. 16 shows the pressure and torque coefficients for the tilt angles $\beta = 5^\circ, 0^\circ$, and -5° under different slopes. The pressure coefficient and torque coefficient show similar variation trends. When the slope is large, the pressure coefficient and torque coefficient become large. In practical engineering, 5° will be selected as the wind protection angle, so that when PV modules are installed on the hillside, the strategy should be changed appropriately.

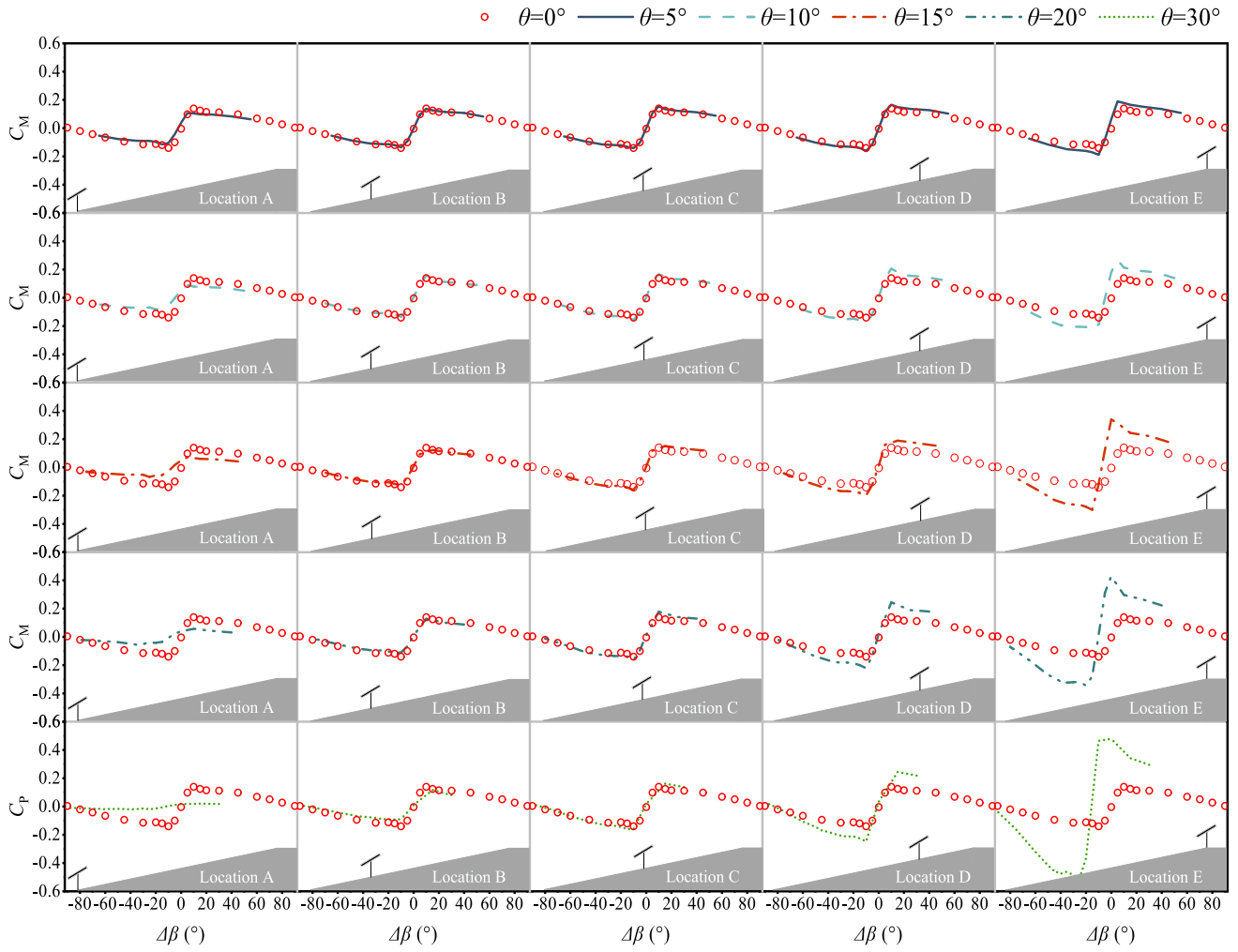


Fig. 13. Comparison of torque coefficients of single-row PV modules at different locations on the hillside and on the flat ground.

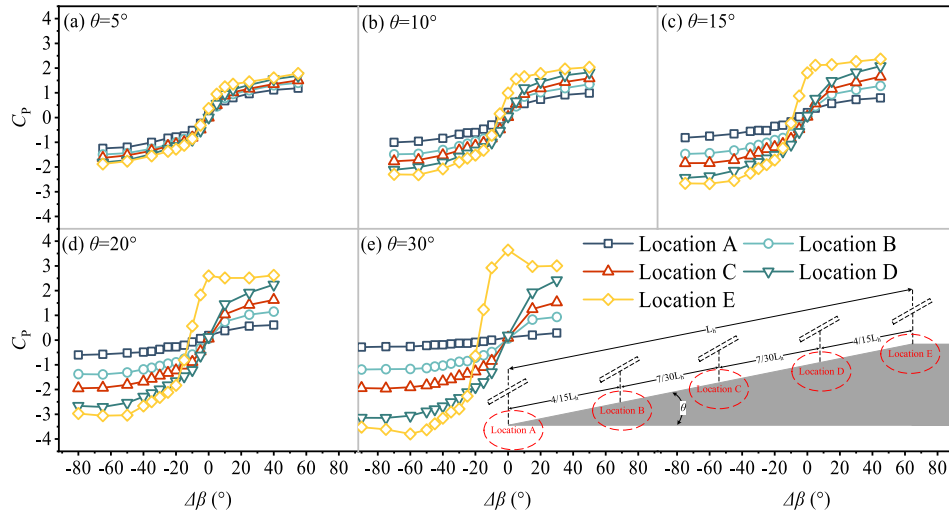


Fig. 14. Pressure coefficients of single-row PV modules under different slopes.

3.3. Wind load analysis of the PV array

Fig. 17 illustrates the location of the PV array, showing eight rows of PV modules named R1 to R8 along the hillside, with row spacing denoted as s . The PV array is located precisely between the previously

mentioned Location B and Location D, where the wind direction is almost parallel to the hillside and relatively stable, avoiding the disturbance of the flow field by the bottom and top of the hillside.

Fig. 18 displays the pressure coefficients for the PV modules at various array locations. The figure reveals a significant shielding effect

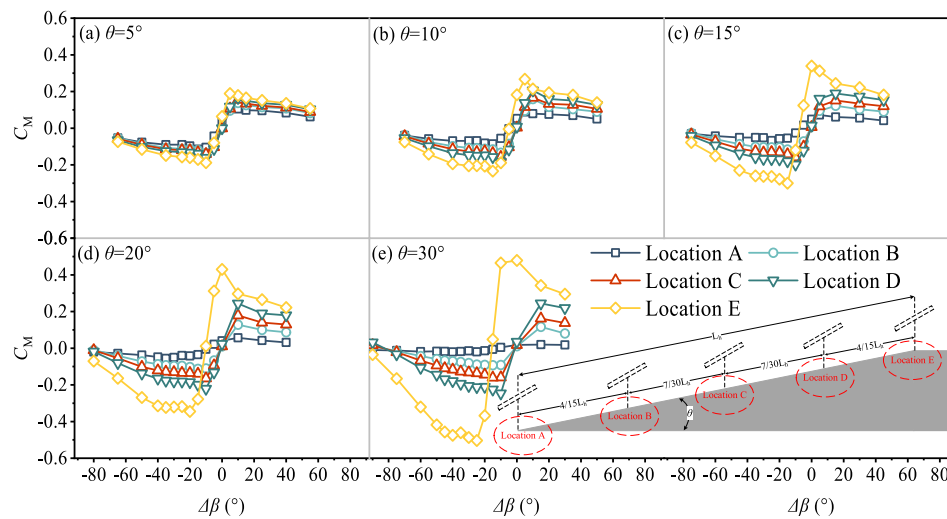


Fig. 15. Torque coefficients of single-row PV modules under different slopes.

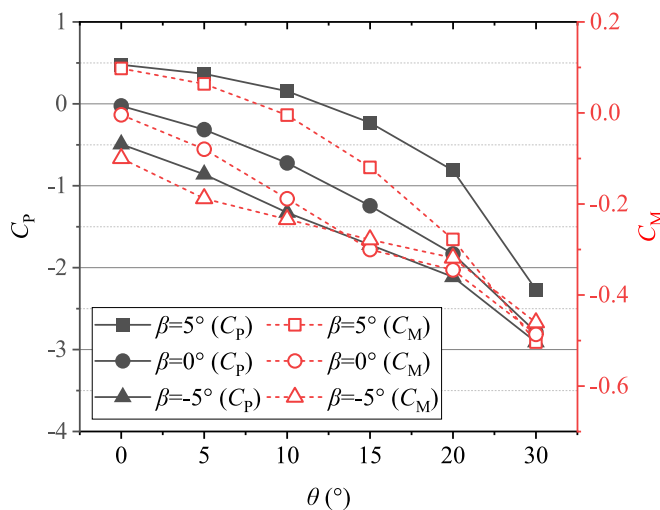


Fig. 16. Pressure coefficient and torque coefficient of tilt angle $\beta = 5^\circ, 0^\circ, -5^\circ$ under different slopes.

within the hillside PV array, which increases with larger relative tilt angles of the PV modules. Similarly to R0, the pressure coefficient R1 varies with the relative tilt angle. The pressure coefficients R2–R8 follow a similar trend with respect to the relative tilt angle, peaking near $\pm 15^\circ$. The enhanced shielding effect at larger relative tilt angles results in lower pressure coefficients for R2–R8. With decreasing relative tilt angle, the shielding effect diminishes, and the windward area of the PV

modules decreases. This combined effect leads to the pressure coefficients of R2–R8 peaking near $\pm 15^\circ$.

The impact of slope on the pressure coefficients differs between R1 and R2–R8. For R1, a steeper slope results in a smaller pressure coefficient due to its weakening influence on the pressure at this location; this effect is enhanced with greater slope steepness. For R2–R8, a steeper slope leads to higher pressure coefficients, which is attributed to the significant impact of the slope on weakening the shielding effect of the PV array.

Fig. 19 displays the torque coefficients of the PV modules at various locations in the array. Similar to that of R0, the torque coefficient of R1 varies with the relative tilt angle, peaking near $\pm 15^\circ$. This is due to the combined effects of the surface pressure and its distribution: larger relative tilt angles induce greater but more evenly distributed pressure, reducing the torque coefficient; smaller angles give rise to an uneven pressure distribution but lower pressure, also reducing the torque coefficient. The impact of the slope on the torque coefficients is similar to its effect on pressure coefficients: for R1, larger slopes reduce the torque coefficient; for R2–R8, the torque coefficient is increased by larger slopes.

Fig. 20 compares the pressure coefficients of each row of the modules in a PV array under different slopes, with a significant shielding effect exerted on the back row by the front row; the shielding effect is enhanced with an increase in the relative tilt angle. The pattern of the changes in the pressure coefficients for PV modules at different locations within the array varies. For R1, the pressure coefficient increases with increasing relative tilt angle. For R2–R8, the peak pressure coefficients are found near $\pm 15^\circ$ relative tilt angles. This is due to both the shielding effect of the front row and the windward area of the PV modules. When the relative tilt angle is too large, the windward area of the PV modules

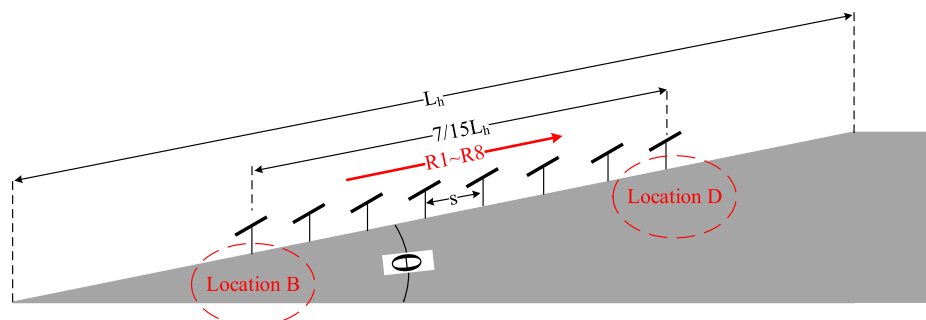


Fig. 17. Location diagram of PV array.

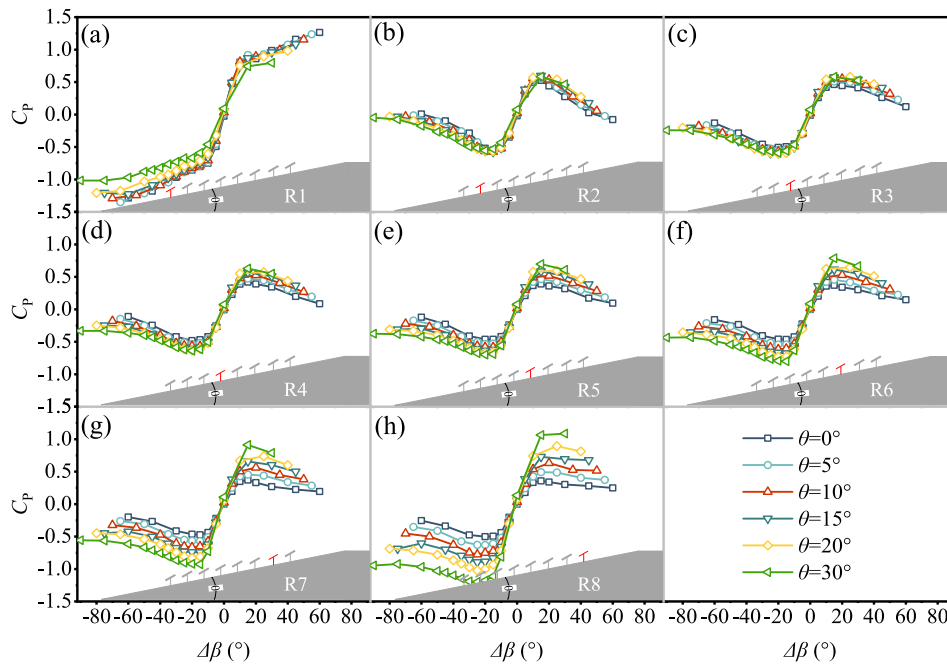


Fig. 18. Pressure coefficients of PV modules at different locations in the array.

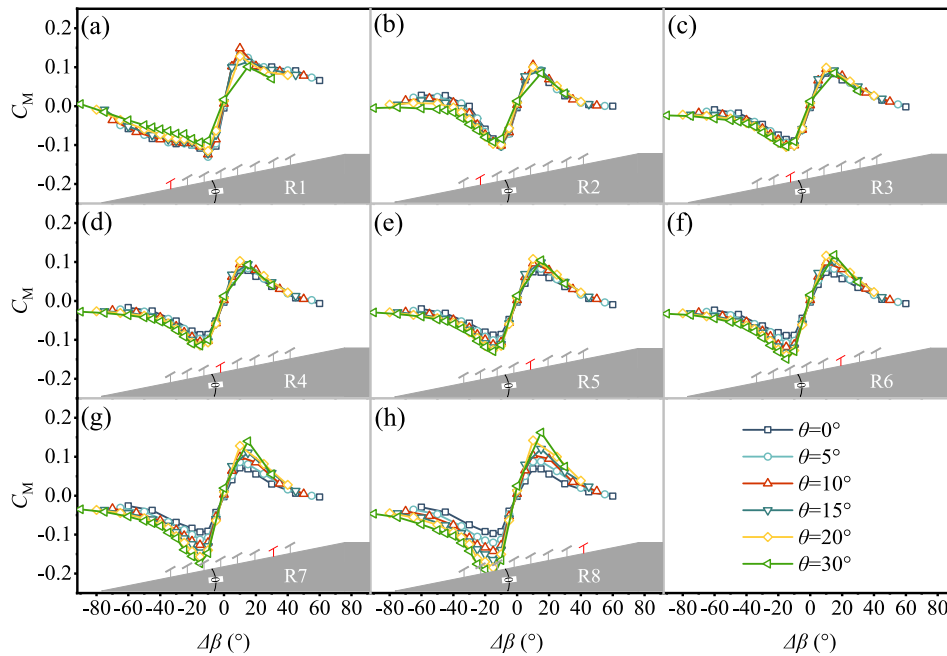


Fig. 19. Torque coefficients of PV modules at different locations in the array.

is large, but so is the shielding effect from the front row; when the relative tilt angle is too small, the shielding effect from the front row is lower, but the windward area of the PV modules is also small, both of which lead to a smaller pressure coefficient. R2 is the location that is most strongly affected by the shielding effect, with the shielding effect on R3–R8 gradually weakening with greater separation from the front row. As the slope increases, the shielding effect gradually weakens.

At $\theta = 0^\circ, 5^\circ$, and 10° , the pressure coefficients of R2–R8 are lower than that of R1. At $\theta = 15^\circ, 20^\circ$, and 30° , the pressure coefficients of R2–R8 exceed that of R1 at smaller angles, as highlighted by the red rectangle in the figure.

Fig. 21 shows a comparison of the torque coefficients for each row of

the modules in a PV array under different slopes; a significant shielding effect from the front row on the back row is evident, and the shielding effect is enhanced with increasing relative tilt angle. Torque coefficients arise from both surface pressure on the PV modules and an uneven pressure distribution. The torque coefficients at various array locations follow a similar pattern, peaking near relative tilt angles of $\pm 15^\circ$, with very large or small angles yielding lower coefficients. This trend is observed because larger relative tilt angles increase the surface pressure with a uniform distribution, while smaller angles lead to an uneven distribution but lower overall pressure. Both scenarios lead to reduced torque coefficients for PV modules. As the slope increases, the shielding effect gradually weakens. When $\theta = 0^\circ$ and 5° , the torque coefficients of

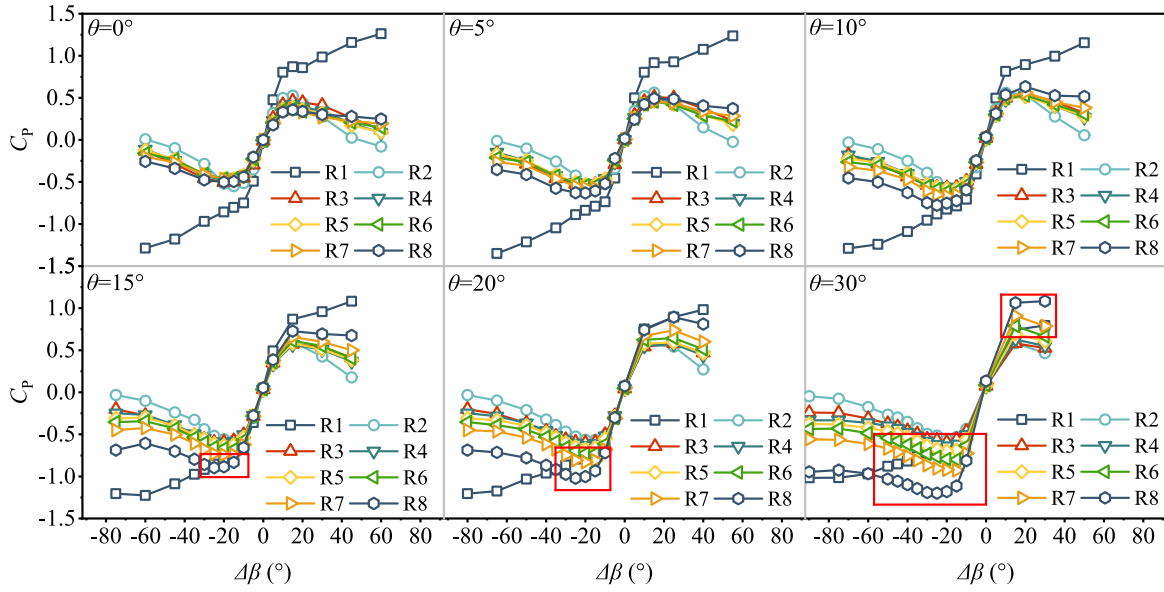


Fig. 20. Comparison of pressure coefficients of each row of modules in PV arrays under different slopes.

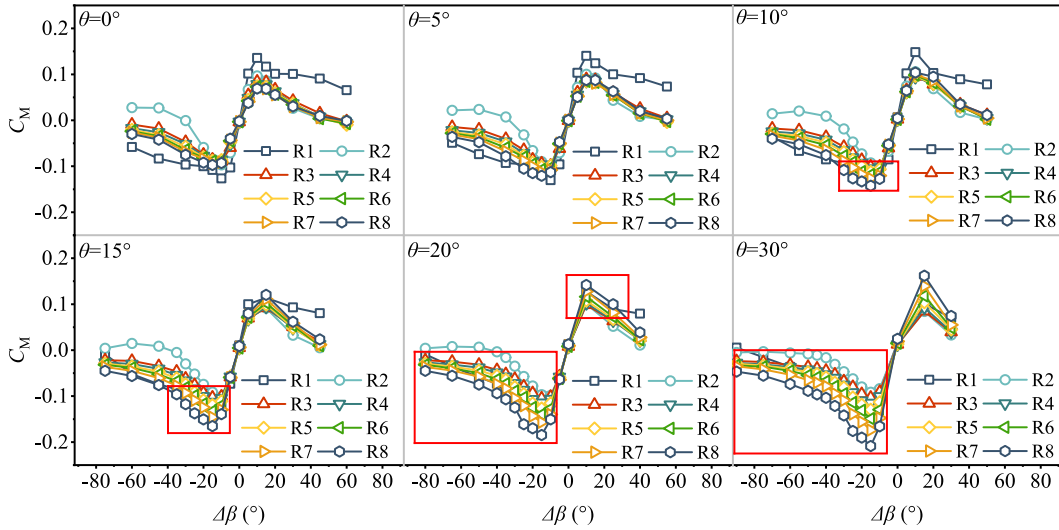


Fig. 21. Comparison of torque coefficients of each row of modules in PV arrays under different slopes.

R2–R8 are less than that of R1. At $\theta = 0^\circ$ and 5° , the torque coefficients of R2–R8 are lower than that of R1. At $\theta = 10^\circ, 15^\circ, 20^\circ$, and 30° , the torque coefficients of R2–R8 exceed that of R1 at specific angles, highlighted by the red area in the figure.

3.4. Flow field analysis

Flow field analysis revealed that slope significantly affects the wind loads on both single-row PV modules and PV arrays on hillsides. This section analyses the impact of the slope on the wind loads from the perspective of the flow field, revealing the mechanism of the effect of the slope on the PV modules at different locations and on the shielding effect for PV arrays.

Fig. 22 presents the pressure coefficients and velocity contours for single-row PV modules at the $\beta = 30^\circ$ tilt at the bottom of hillsides with varying slopes. The figure clearly shows significant variations in the pressure and velocity distributions around the PV modules due to varying hillside slopes. As shown in Fig. 22 (right), the wind speed and direction at the bottom of the hillside are influenced by the slope. The

blocking effect of the hillside means that steeper slopes lead to more pronounced reductions in wind speed. This leads to the formation of a low-wind-speed area near the bottom of the hillside, with a larger slope resulting in a larger low-wind-speed area and lower wind speeds within the area. Thus, the pressure coefficients of the PV modules in this area vary significantly with the slope. Additionally, hillsides significantly alter the flow field pattern around the PV modules, notably inhibiting vortex development on the leeward side. A steeper slope produces a stronger inhibitory effect. As shown in Fig. 22 (left), at $\theta = 5^\circ$, the blocking effect of the hillside is smaller, the wind speed near the bottom of the hillside is still high, and the wind speed reduction due to PV module obstruction results in a larger wind speed difference near the PV module. Consequently, positive pressure on the windward side and negative pressure on the leeward side result in a higher net wind pressure coefficient on the PV module. At $\theta = 15^\circ$, the hillside's increased blocking effect lowers the wind speeds near the bottom, diminishing the wind speed difference near the PV module compared to that at $\theta = 5^\circ$. At this time, both the windward and leeward sides of the PV module experience positive pressure, resulting in a smaller wind pressure

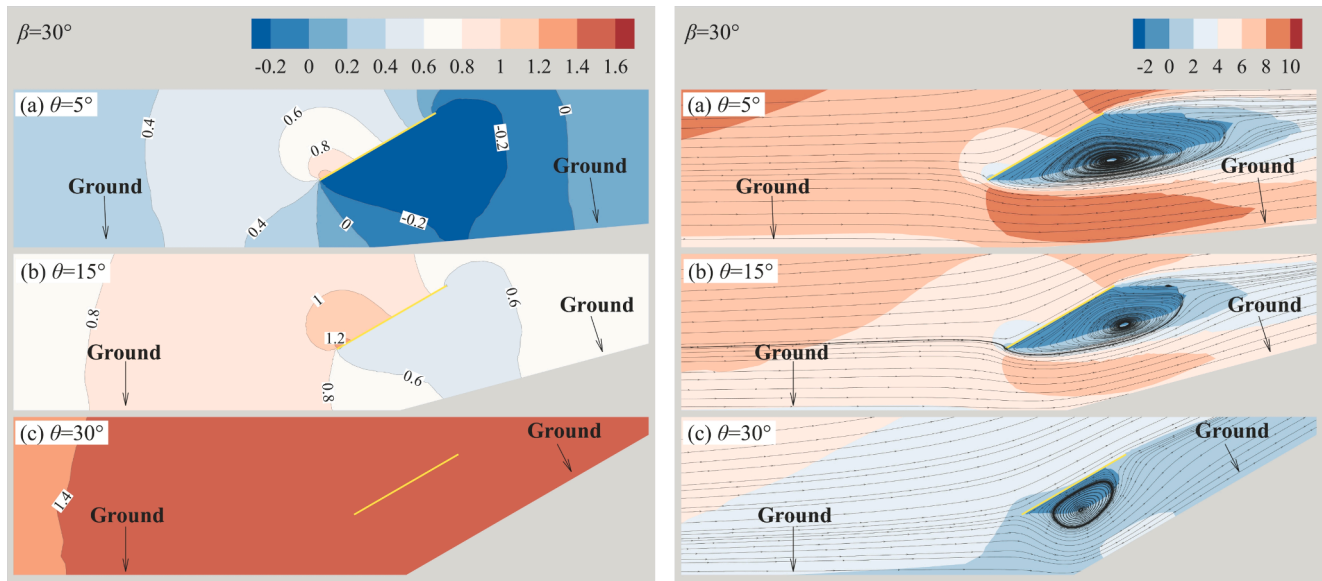


Fig. 22. Pressure coefficient contour map (left) and velocity contour map (right) of single-row PV modules at the bottom of hillsides with different slopes ($y_i/L_h = 0$, $\beta = 30^\circ$).

coefficient on the PV module. At $\theta = 30^\circ$, the blocking effect of the hillside further increases, drastically lowering the wind speed at the bottom, with a minimal wind speed difference occurring near the PV module. Consequently, positive pressure on both sides of the PV module, with almost no pressure difference, renders the net pressure coefficient nearly 0.

Fig. 23 presents the pressure coefficient and velocity contour maps for single-row PV modules at a $\beta = 30^\circ$ tilt, situated mid-hillside under various slopes. The incoming wind is affected by the hillside, and the wind direction at the bottom of the hillside changes. The wind direction at the middle of the hillside is almost parallel to the hillside, as shown in Fig. 23 (right). The wind is affected by the PV module, and the wind speed difference is generated between the upper edge and the lower edge, thereby generating a vortex on the back row and generating a negative pressure zone. With increasing $\Delta\beta$, the scale of the vortex, the range of the negative pressure zone, and the net pressure coefficient of

the PV module increase. That is, the relative tilt angle in the middle of the hillside is the main factor affecting the flow field distribution and the net pressure coefficient of the PV module.

Fig. 24 shows the pressure coefficient and velocity contour maps for single-row PV modules at a tilt angle of $\beta = 30^\circ$ located at the top of hillsides under various slopes. Comparison of Figs. 22 and 24 reveals that while turning angles influence the flow fields at both hillside ends, the effects are different: steeper slopes reduce wind speed gradients at the bottom, while increasing them at the top. Consequently, increasing the slope decreases the pressure gradient across the PV modules at the bottom of the hillside and increases it at the top of the hillside. At the top of the hillside, the flow field is affected not only by the PV module but also by the turning angle. The wind direction gradually changes from parallel to the hillside in the middle of the hillside to parallel to the horizontal plane; that is, the wind gradually recovers to the incoming wind direction. The flow field near the PV module is affected by the

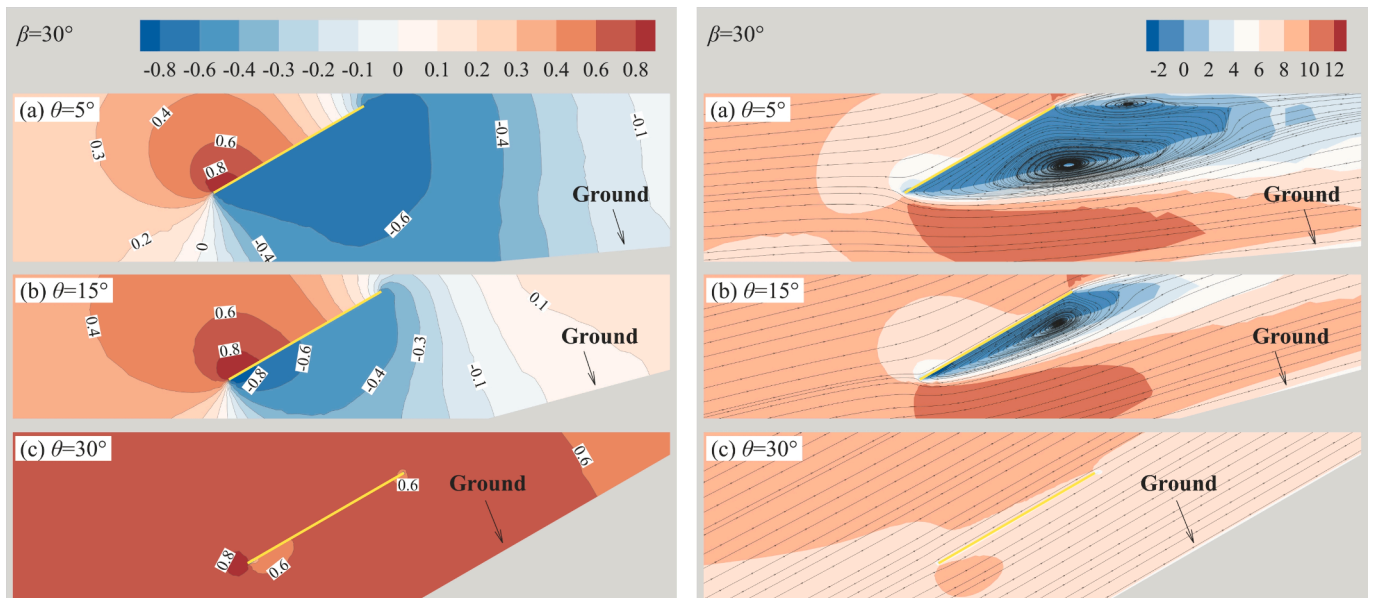


Fig. 23. Pressure coefficient contour map (left) and velocity contour map (right) of single-row PV modules at the middle of hillsides with different slopes ($y_i/L_h = 0.5$, $\beta = 30^\circ$).

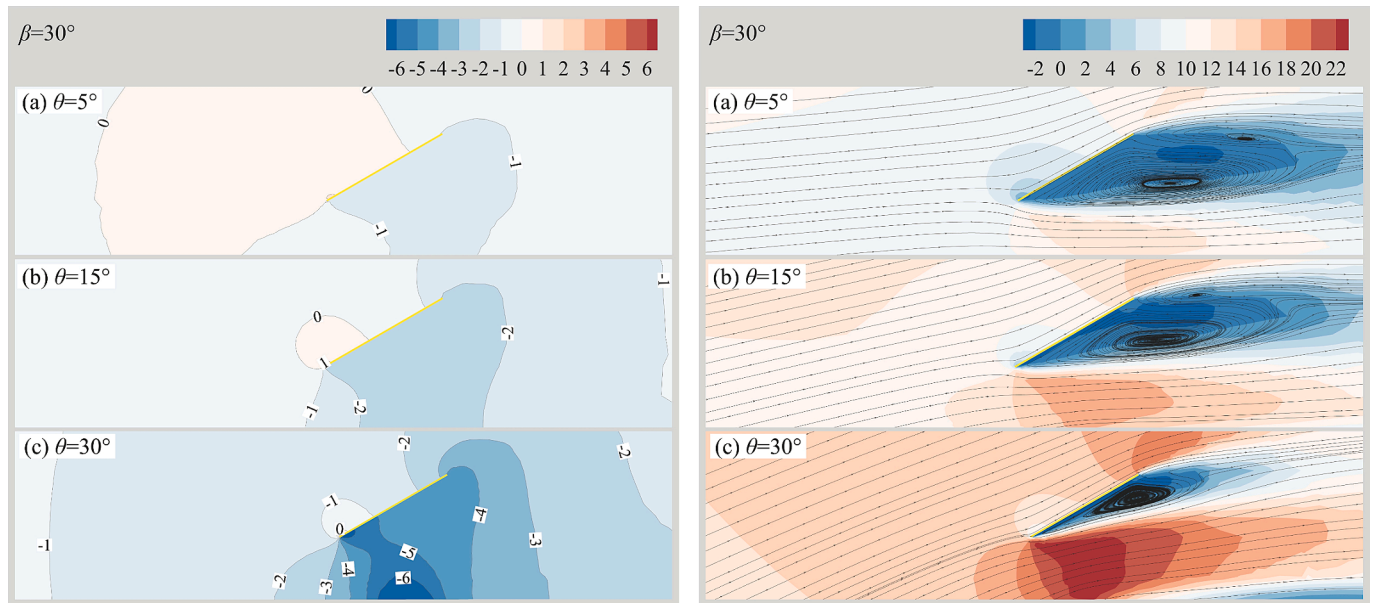


Fig. 24. Pressure coefficient contour map (left) and velocity contour map (right) of single-row PV modules at the top of hillsides with different slopes ($y_t/L_h = 1$, $\beta = 30^\circ$).

relative tilt angle and the turning angle. The flow fields for the slopes of 5° and 15° are not very different, but when the slope is 30° , the flow field is significantly different, possibly due to the flow separation phenomenon at the top of the hillside. The PV array has an evident shielding effect, which is weakened by decreasing the tilt angle. When the PV array is located on the hillside, the shielding effect will also be affected by the slope.

Fig. 25 presents the pressure coefficients and velocity contours for the first three rows (R1, R2, R3) of a PV array at a $\beta = 30^\circ$ tilt across various slopes. The flow field around the PV array is unaffected by the slope turning angle, as the wind direction remains parallel to the hillside and is primarily influenced by the PV modules. The PV array has an evident shielding effect, which is weakened by decreasing the tilt angle. When the PV array is located on the hillside, the shielding effect will also be affected by the slope. Fig. 25 (left) illustrates that the slope significantly affects the shielding effect on the PV array, which weakens with

increasing slope. As shown in Fig. 25 (right), the scale of the vortices on the leeward side of the PV modules decreases with larger slope, leading to a reduction in the negative pressure area on the leeward side, which is similar to the pattern of the changes with the tilt angle for the PV arrays on flat ground.

Fig. 26 displays the pressure coefficients and velocity contours for the first three rows (R1, R2, R3) of the PV arrays at the same relative tilt angle $\Delta\beta$ across various slopes. It is evident from the figure that similar relative tilt angles $\Delta\beta$ result in similar pressure and velocity distributions, although the numerical values of the pressure and velocity differ. This is because the wind speed varies at the different locations on the hillside, and the blocking effect of the hillside leads to lower wind speeds at the bottom, with the wind speed gradually recovering and accelerating from the bottom to the top of the hillside. As shown in Fig. 26 (right), when $\theta = 15^\circ$, R1, R2, and R3 are located in the wind speed recovery zone, where the wind speed has not fully recovered and is

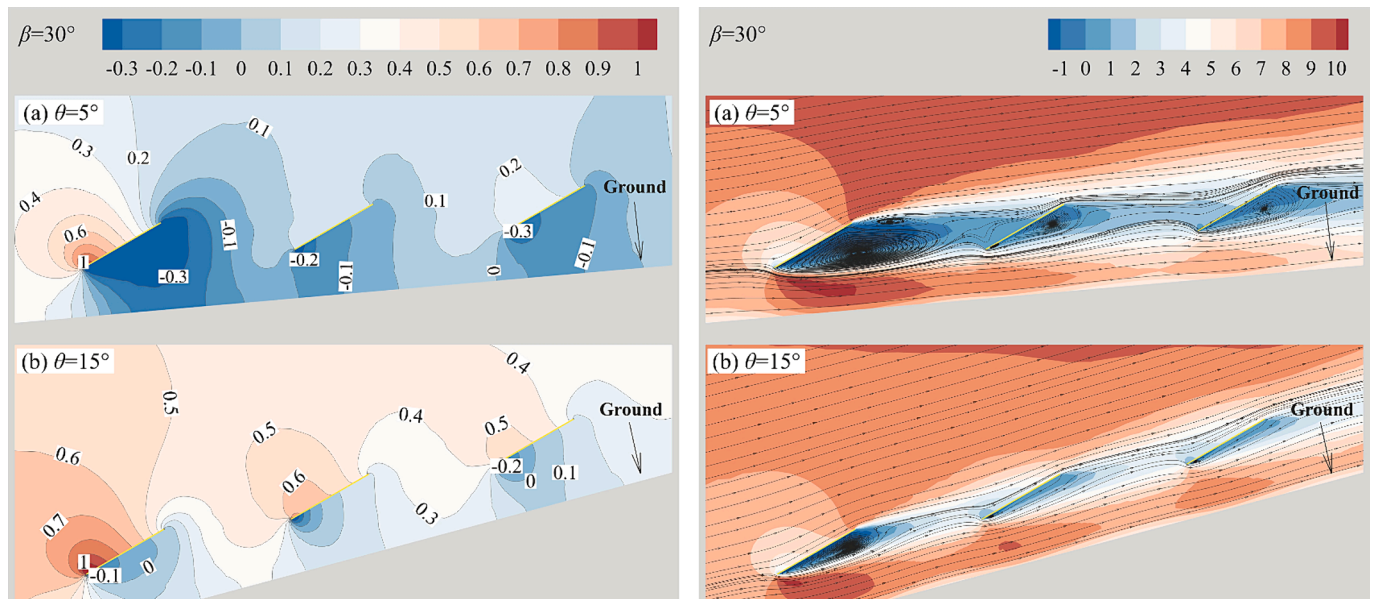


Fig. 25. Pressure coefficient contour map (left) and velocity contour map (right) for R1, R2, and R3 ($\beta = 30^\circ$) under different slopes.

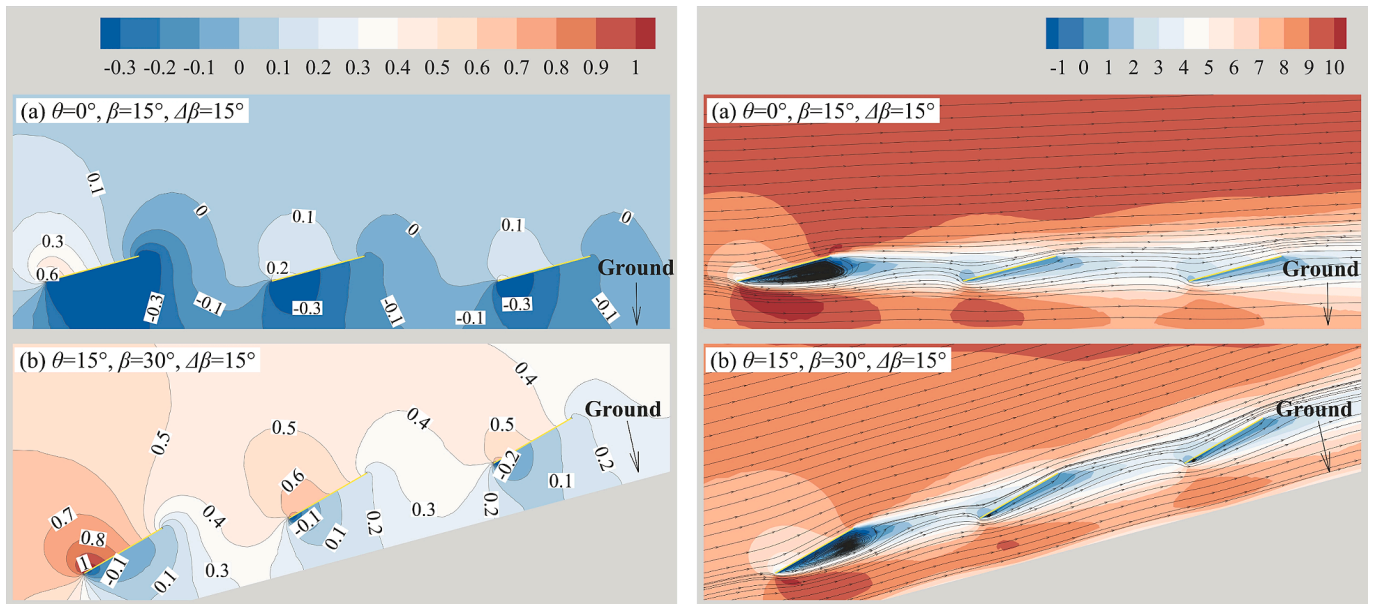


Fig. 26. Pressure coefficient contour map (left) and velocity contour map (right) for R1, R2, and R3 with the same relative tilt angles ($\beta = 15^\circ$, $\beta = 30^\circ$) under different slopes.

lower than the wind speed at the same height above the ground when $\theta = 0^\circ$.

4. Conclusion

This paper studies the wind loads and mechanism of single-row PV modules and PV arrays on hillsides and the following conclusions are reached:

- (1) The relative tilt angle, $\Delta\beta$, significantly influences the wind load on PV modules. The relative tilt angle provides a more accurate description of the wind load changes on PV modules on hillsides than the tilt angle, β .
- (2) Wind direction undergoes significant changes at both the top and bottom of a hillside. The angle between the wind direction and the hillside, both at the top and bottom, is consistently approximately half of the slope's angle, represented as $\Delta\theta \approx 0.5\theta$.
- (3) The impact of the slope on the wind load on PV modules varies by location, transitioning from the weakening effect at the bottom to the enhancement at the top of the hillside. The wind load of the PV modules in the range of $0.27L_h$ from the bottom of the hillside is generally lower than that of the ground PV modules. When the slope is 30° , the wind load of the PV modules at the bottom of the hillside is approximately 20 % of that of the ground. The wind load of the PV modules in the range of $0.27L_h \sim 0.5L_h$ from the bottom of the hillside is not much different from that of the ground PV modules. The wind load of the PV modules in the range of $0.5L_h \sim L_h$ from the bottom of the hillside is greater than that of the ground PV modules. When the slope is 30° , the wind load of the PV modules at the top of the hillside is approximately 250 % of that of the ground.
- (4) According to the monitoring results of the wind speed on Line I and Line II, the PV in the range of $0.2L_h$ from the top of the hillside should receive special attention during the design process. In this range, the wind direction shows clear changes, and the acceleration effect of the wind speed is severe. The acceleration effect becomes stronger with greater slope in this range. At a slope of 30° , the maximum streamwise wind speed ratio reaches 1.85, and the vertical wind speed ratio reaches 0.8. In this range, the wind speed parallel to the flow or vertical at the lower edge of

the PV module surpasses that at the upper edge, leading to a unique wind load pattern distinct from that of flat ground or other hillside areas.

- (5) There is a significant shielding effect in the PV arrays on hillsides, with the strength of the shielding effect decreasing with greater slope. For the PV array placement in this paper, the slope has a weakening effect on the wind load of R1. The weakening effect becomes stronger with larger slope, and the maximum wind load can be reduced by 25 %. The wind load on R2–R8 shows an enhancement effect. The enhancement effect increases with larger slope, and the maximum enhancement for R8 reaches 380 %. This is because a greater slope leads to a weaker shielding effect and a stronger acceleration effect. In the process of structural design, if the slope is large and the relative tilt angle is near $\pm 20^\circ$, R2–R8 should receive more attention. In this case, the pressure coefficient and torque coefficient of R2–R8 may exceed those of R1.
- (6) When the relative tilt angles are the same, the patterns of pressure and wind speed around the PV modules are similar, yet their values vary due to the wind acceleration effect. Consequently, the wind load on the single-row PV modules on hillsides arises from both the relative tilt angle and terrain-induced wind acceleration. Similarly, the wind load on the PV arrays on hillsides is determined by the interplay of shielding and terrain acceleration.

This paper still has some shortcomings. First, the model used in this paper is a two-dimensional hillside. Typical three-dimensional slopes and real complex mountains will be investigated in the follow-up research. Second, according to previous experience, there is no turbulence model that can fully simulate the real state of the system. The conclusion of this paper is based solely on the results of CFD numerical simulations, which have certain limitations. Finally, wind tunnel tests and field measurements are needed for further verification.

CRediT authorship contribution statement

Ang Xu: Writing – original draft, Visualization, Software, Data curation. **Wenyong Ma:** Methodology, Funding acquisition. **Wenshuo Yang:** Writing – review & editing. **Mingyang Li:** Writing – review & editing. **Yongjie Tan:** Investigation. **Chuang Zou:** Investigation.

Shengguan Qiang: Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] A. Abiola-Ogedengbe, H. Hangan, K. Siddiqui, Experimental investigation of wind effects on a standalone photovoltaic (PV) module, *Renew. Energy* 78 (2015) 657–665.
- [2] A. Agarwall, H. Irtaza, K. Shahab, Aerodynamic wind pressure on solar PV arrays mounted on industrial pitched roof building, *Proceedings of the International Conference on Recent Advances in Engineering & Science (ICRAES-2020)*, Aligarh, India, 2020, pp. 4–6.
- [3] A.M. Aly, On the evaluation of wind loads on solar panels: The scale issue, *Sol. Energy* 135 (2016) 423–434.
- [4] A. Bahrami, C.O. Okoye, U. Atikol, Technical and economic assessment of fixed, single and dual-axis tracking PV panels in low latitude countries, *Renew. Energy* 113 (2017) 563–579.
- [5] M.T.L. Browne, Z.J. Taylor, S. Li, S. Gamble, A wind load design method for ground-mounted multi-row solar arrays based on a compilation of wind tunnel experiments, *J. Wind Eng. Ind. Aerodyn.* 205 (2020) 104294.
- [6] P. Fang, D. Zheng, L. Li, W. Ma, S. Tang, Numerical and experimental study of the aerodynamic characteristics around two-dimensional terrain with different slope angles, *Frontiers of Earth Science* 13 (4) (2019) 705–720.
- [7] P. Fang, D. Zheng, L. Li, W. Ma, S. Tang, Numerical and experimental study of the aerodynamic characteristics around two-dimensional terrain with different slope angles, *Frontiers of Earth Science* 13 (2019) 705–720.
- [8] C. Hu, L. Zhao, Y. Ge, Time-frequency evolutionary characteristics of aerodynamic forces around a streamlined closed-box girder during vortex-induced vibration, *J. Wind Eng. Ind. Aerodyn.* 182 (2018) 330–343.
- [9] H. Irtaza, A. Agarwal, CFD Simulation of Turbulent Wind Effect on an Array of Ground-Mounted Solar PV Panels, *Journal of the Institution of Engineers (India): Series A* 99 (2) (2018) 205–218.
- [10] C.M. Jubayer, H. Hangan, Numerical simulation of wind effects on a stand-alone ground mounted photovoltaic (PV) system, *J. Wind Eng. Ind. Aerodyn.* 134 (2014) 56–64.
- [11] C.M. Jubayer, H. Hangan, A numerical approach to the investigation of wind loading on an array of ground mounted solar photovoltaic (PV) panels, *J. Wind Eng. Ind. Aerodyn.* 153 (2016) 60–70.
- [12] K. Kondo, M. Tsuchiya, S. Sanada, Evaluation of effect of micro-topography on design wind velocity, *J. Wind Eng. Ind. Aerodyn.* 90 (12–15) (2002) 1707–1718.
- [13] M. Koussa, A. Cheknane, S. Hadji, M. Haddadi, S. Noureddine, Measured and modelled improvement in solar energy yield from flat plate photovoltaic systems utilizing different tracking systems and under a range of environmental conditions, *Appl. Energy* 88 (5) (2011) 1756–1771.
- [14] T. Kray, D. Markus, Peak Wind Loads on Single-axis PV Tracking Systems. The 15th International Conference on Wind Engineering, 2019.
- [15] M. Lave, J. Kleissl, Optimum fixed orientations and benefits of tracking for capturing solar radiation in the continental United States, *Renew. Energy* 36 (3) (2011) 1145–1152.
- [16] L. Li, D. Zheng, G. Chen, P. Fang, W. Ma, S. Tang, Large eddy simulation of flow over a three-dimensional hill with different slope angles, *Frontiers of Earth Science* (2023).
- [17] Y.F. Lun, A. Mochida, H. Yoshino, S. Murakami, Applicability of linear type revised k- ϵ models to flow over topographic features, *J. Wind Eng. Ind. Aerodyn.* 95 (5) (2007) 371–384.
- [18] W. Ma, W. Zhang, X. Zhang, W. Chen, Q. Tan, Experimental investigations on the wind load interference effects of single-axis solar tracker arrays, *Renew. Energy* 202 (2023) 566–580.
- [19] E. Martínez-García, E. Blanco-Marigorta, J.P. Gayo, A. Navarro-Manso, Influence of inertia and aspect ratio on the torsional galloping of single-axis solar trackers, *Eng. Struct.* 243 (2021) 112682.
- [20] E. Martínez García, E. Blanco Marigorta, J.L. Parrondo Gayo, A. Navarro Manso, Experimental determination of the resistance of a single-axis solar tracker to torsional galloping, *Struct. Eng. Mech.* (2021).
- [21] Miller, R.D., Zimmerman, D.K., 1981. Wind loads on flat plate photovoltaic array fields. Phase III, final report. United States, p. Medium: ED; Size: Pages: 383.
- [22] G.P. Reina, G. De Stefano, Computational evaluation of wind loads on sun-tracking ground-mounted photovoltaic panel arrays, *J. Wind Eng. Ind. Aerodyn.* 170 (2017) 283–293.
- [23] P. Richards, R. Hoxey, Appropriate boundary conditions for computational wind engineering models using the k- ϵ turbulence model, *J. Wind Eng. Ind. Aerodyn.* 46 (1993) 145–153.
- [24] C. Rohr P.A. Bourke D. Banks Torsional Instability of Single-Axis Solar Tracking Systems 2015 Porto Alegre 21 26.
- [25] M. Shademan, R. Balachandar, R.M. Barron, Detached eddy simulation of flow past an isolated inclined solar panel, *J. Fluids Struct.* 50 (2014) 217–230.
- [26] M. Shademan, R.M. Barron, R. Balachandar, H. Hangan, Numerical simulation of wind loading on ground-mounted solar panels at different flow configurations, *Can. J. Civ. Eng.* 41 (8) (2014) 728–738.
- [27] T.-H. Shih, W.W. Liou, A. Shabbir, Z. Yang, J. Zhu, A new k- ϵ eddy viscosity model for high reynolds number turbulent flows, *Comput. Fluids* 24 (3) (1995) 227–238.
- [28] Y. Tominaga, A. Mochida, R. Yoshie, H. Kataoka, T. Nozu, M. Yoshikawa, T. Shirasawa, ALJ guidelines for practical applications of CFD to pedestrian wind environment around buildings, *J. Wind Eng. Ind. Aerodyn.* 96 (10–11) (2008) 1749–1761.
- [29] D. Valentin, C. Valero, M. Egusquiza, A. Presas, Failure investigation of a solar tracker due to wind-induced torsional galloping, *Eng. Fail. Anal.* 135 (2022) 106137.
- [30] J.P. Van Doormaal, G.D. Raithby, Enhancements of the SIMPLE method for predicting incompressible fluid flows, *Numerical Heat Transfer* 7 (2) (1984) 147–163.
- [31] R.G. Vieira, F.K.O.M.V. Guerra, M.R.B.G. Vale, M.M. Araújo, Comparative performance analysis between static solar panels and single-axis tracking system on a hot climate region near to the equator, *Renew. Sustain. Energy Rev.* 64 (2016) 672–681.
- [32] W.P. Warsido, G.T. Bitsuamlak, J. Barata, A. Gan Chowdhury, Influence of spacing parameters on the wind loading of solar array, *J. Fluids Struct.* 48 (2014) 295–315.
- [33] A. Xu, W. Ma, H. Yuan, L. Lu, The effects of row spacing and ground clearance on the wind load of photovoltaic (PV) arrays, *Renew. Energy* 220 (2024) 119627.
- [34] Q. Yang, T. Zhou, B. Yan, M. Liu, P. Van Phuc, Z. Shu, LES study of topographical effects of simplified 3D hills with different slopes on ABL flows considering terrain exposure conditions, *J. Wind Eng. Ind. Aerodyn.* 210 (2021).
- [35] Y. Yang, M. Gu, S. Chen, X. Jin, New inflow boundary conditions for modelling the neutral equilibrium atmospheric boundary layer in computational wind engineering, *J. Wind Eng. Ind. Aerodyn.* 97 (2) (2009) 88–95.